

Rank Order Is Mostly Noise

Confidence intervals for brand visibility in AI answer engines, and a failure of the citation-extraction layer

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Abstract

Vendors and agencies increasingly publish ordered leaderboards of which brands large language models recommend, almost always as bare rank orders without any statement of uncertainty. We measured 4,497 answers across 9 B2B software categories and 90 products, using 5 AI engines and 10 pre-registered buyer prompts at 10 runs per prompt per engine, and attached a 95% confidence interval to every share estimate. **In 6 of 9 categories the first-ranked product is not statistically separable from the second.** In the most extreme case the top four products are mutually indistinguishable. Category leaders cluster in a narrow band of 16.4% to 25.0% share of model, suggesting a ceiling on how dominant any single brand becomes in a competitive category. We conclude that published rank orders in this field convey less information than they appear to, and that reporting a rank without an interval is not a presentational shortcut but a substantive overstatement.

We additionally report a negative methodological result. Our own citation-extraction layer failed unevenly across engines: one engine returned **zero** distinct cited domains in all 9 categories, a second returned 1–2 against a list cap of 5, and 39% of extracted citations resolved to a platform redirect wrapper rather than a source. We therefore withdraw the citation-level findings of an earlier draft and report the extraction failure itself, because it has a consequence beyond this paper: citation-share figures are published across this sector without any disclosure of extraction quality, and our results indicate that quality cannot be assumed.

Keywords: generative engine optimization, answer engine optimization, large language models, brand visibility, measurement error, confidence intervals, pre-registration, citation extraction

Note on this version

Version 1.0 of this paper reported a correlation between how often a product is named and how often its domain is cited ($r = +0.32$), and a categorical spread in citation rates. Both rested on citation data we subsequently found to be defective, and both are withdrawn here. The share-of-model findings, which derive from brand naming and not from citations, are unchanged and independently reproducible from the released data. Section 4.4 sets out the defect in full.

1. Introduction

A new measurement practice has formed around the question of which brands large language models recommend. Vendors publish visibility indexes, agencies publish category leaderboards, and both typically present an ordered list: first, second, third. The ordering carries an implicit claim, which is that the ordering means something.

That claim is rarely tested. An LLM answering the same prompt twice does not necessarily name the same products, and any share estimate derived from a finite number of sampled answers therefore carries sampling error. Whether the observed gap between rank one and rank two exceeds that error is an empirical question, and

one the field has largely not asked in public.

We ask it here. Across 9 categories, 90 products and 4,497 sampled answers, we compute share of model with an explicit interval and test whether adjacent ranks are separable. They usually are not. We also report, as a distinct contribution, a failure in our own instrument: the layer that extracts cited sources from answers does not work uniformly across engines, and we could not have known that from any published account of how these measurements are made, because no such account exists.

2. The state of practice

Three properties characterise most publicly available measurement in this area. First, the sampling design is undisclosed: how many prompts, how many repetitions per prompt, and over what window is typically absent. Second, no uncertainty is reported, so a reader cannot distinguish a robust lead from a coin flip. Third, the prompt set and the set of brands considered are usually chosen after the fact, which permits a leaderboard to be composed rather than measured.

A fourth property is less often noticed and is the subject of Section 4.4. Where a study reports how often a brand's own domain is *cited* as a source, it depends on an extraction layer that reads citations out of each engine's response. Engines differ substantially in what they return, and some return links that point at the platform rather than at a source. We are not aware of any published index that reports the reliability of its own extraction step.

We do not claim these properties are universal, and we have deliberately not named individual vendors: the point is structural rather than an accusation about any particular publisher.

3. Method

3.1 Pre-registration

For each category the prompt set and the brand universe were fixed before any measurement was taken, and both were hashed. The published leaderboard is drawn only from the pre-registered universe, so a product cannot be added because it performed well or dropped because it did not. Each category used 10 prompts and 10 brands; the released data carries the prompt-set hash (for example 4021dabe951b2cda) and the brand-universe hash (0d9663ccf64dc073) alongside the results, so the freeze is checkable rather than merely asserted.

3.2 Sampling

Prompts were issued to 5 engines — Perplexity, Google Gemini, ChatGPT (OpenAI), Claude (Anthropic) and Grok (xAI) — at 10 runs per prompt per engine. Aggregated across categories this yields 4,497 answers. Repetition is the load-bearing design choice: a single response per prompt cannot support any statement about variance, and variance is the thing under examination.

We distinguish throughout between an *engine*, a system that generates an answer, and an *answer surface* such as an AI overview embedded in a conventional search results page. Only engines are measured here. The two are not summed.

3.3 Metric

Share of model is the percentage of all product recommendations, across every prompt and engine, that name a given product: when an engine recommends something in this category, how often is it this product? It is a share of a total that is fixed by construction, so a gain for one product is a loss for another — which is what makes it a

competitive measure rather than a vanity one.

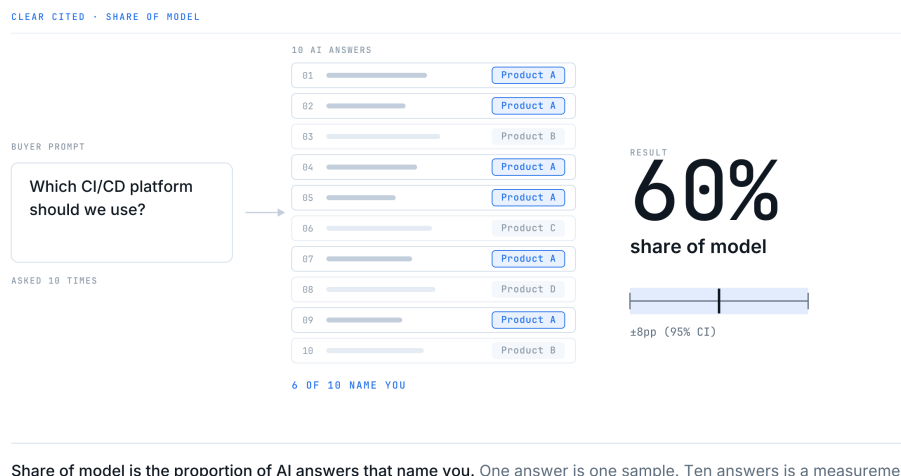


Figure 1. Share of model. Every product mention across all sampled answers is counted; each product's share is its portion of that fixed total, so gains and losses are zero-sum.

3.4 Intervals

Every estimate carries a 95% Wilson score interval ($z = 1.96$). Appearance rate is a proportion over sampled runs, and Wilson is the standard choice for it. Share of model is a ratio over the pooled total of all product mentions in the category, and the same estimator is applied to that denominator. Two products are treated as *not separable* where their intervals overlap.

Wilson's independence assumption does not strictly hold for share of model, and the direction of that error matters. A single answer may name several products, and ten runs of the same prompt are correlated with one another, so the pooled mention denominator contains fewer independent observations than it appears to. The practical effect is that the reported share intervals are *narrower* than correctly clustered intervals would be. A design-effect correction, or a bootstrap resampling answers rather than mentions, would widen them.

This is worth stating precisely because it works against us, not for us. Wider intervals would make *more* adjacent pairs overlap, not fewer. The count of non-separable leaders reported in Section 4.2 is therefore a **lower bound** on how much of the published ranking is indistinguishable from noise, and the paper's central claim is understated rather than overstated by its own estimator.

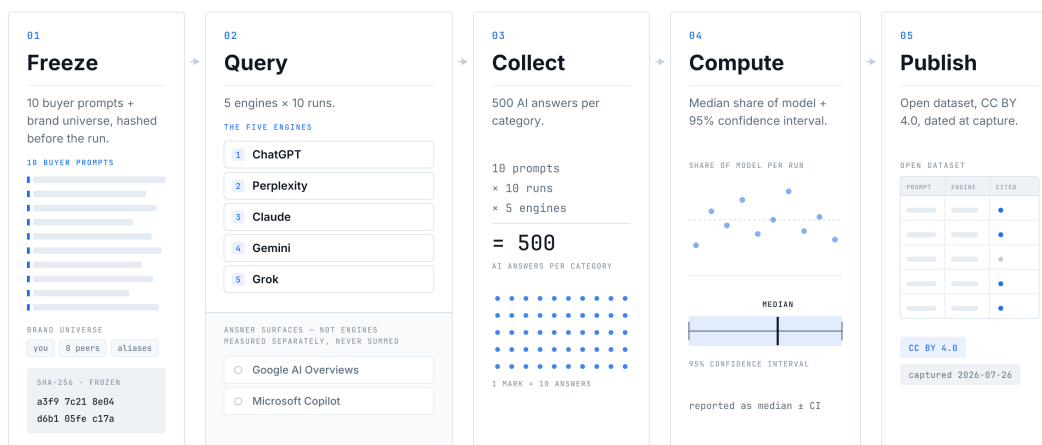
Overlap as a separability criterion is conservative in the opposite direction. Non-overlapping intervals imply a significant difference, but overlapping intervals do not strictly imply the absence of one; a formal test of the difference would reject in some overlapping cases. We adopt overlap because it is the comparison a reader can actually perform from a published table.

3.5 Conflict of interest and exclusions

Clear Cited sells services related to visibility in AI answers. Client domains are excluded from the public ranked index; 6 domains were checked against each category at build time. The policy is disclosed rather than the exclusion hidden. At the time of measurement the business was pre-revenue with no clients, so no exclusion was in fact applied to any category reported here.

How share of model is measured

CLEAR CITED



Clear Cited measurement protocol. Prompts and brand universe frozen before execution.

Figure 2. The measurement pipeline: pre-registered prompts and brand universe, repeated sampling across engines, extraction, aggregation to share of model, and interval estimation. Answer surfaces are measured separately, never summed into the engine count.

4. Results

4.1 Category leaders cluster in a narrow band

Across all 9 categories the top-ranked product's share of model falls between 16.4% and 25.0%. No leader approaches dominance. The narrowness of this band across categories as different as developer CI/CD tooling and CRM software is itself the finding: it suggests engines distribute recommendations across a consideration set rather than converging on a single answer, and that there is a practical ceiling on how much of a competitive category any one brand can occupy.

Category	Leader	Share	95% CI	Top-n indistinguishable
AI observability tools	Datadog	16.4%	15.0–17.9%	4 ■
API platforms	Kong	20.3%	18.6–22.1%	2 ■
CI/CD platforms	GitHub Actions	21.8%	20.1–23.7%	3 ■
CRM software	HubSpot	22.3%	20.6–24.1%	1
Databases	PostgreSQL	22.3%	20.5–24.1%	1
Feature flag platforms	LaunchDarkly	17.2%	15.7–18.8%	2 ■
Incident management platforms	PagerDuty	25.0%	23.1–27.1%	1
Product analytics platforms	Mixpanel	23.2%	21.4–25.1%	3 ■
Vector databases	Weaviate	19.1%	17.6–20.8%	3 ■

Table 1. Category leaders, their share of model with 95% confidence intervals, and how far down the ranking the block of statistically indistinguishable products extends. A value above 1 means the leader is not separable from the products beneath it.

4.2 Rank 1 is not separable from rank 2 in six of nine categories

This is the paper's central result. In 6 of 9 categories the confidence interval of the first-ranked product overlaps that of the second. The published ordering in those categories reflects a difference our sampling cannot distinguish from noise.

The problem is not confined to the top pair. In **AI observability tools** the top 4, **CI/CD platforms** the top 3, **product analytics platforms** the top 3, **vector databases** the top 3 products are mutually indistinguishable. A reader given only the ordered list would reasonably infer a hierarchy among them. There is no evidence for one.

The practical consequence is asymmetric and worth stating plainly. A brand ranked fourth in a block of four indistinguishable products may be told it is losing, and may spend accordingly, when the measurement supports no such conclusion. The same table read with intervals supports a different and more useful statement: this product is in the leading group, and the gap to the leading group's boundary is what would need to move.

4.3 Two edge cases the interval view makes legible

The invisible product. Heatmap, a pre-registered member of the product analytics platforms universe, was named in **zero** of 500 sampled answers. Its share is not small; it is absent. This is a qualitatively different position from ranking last with a low share, and it is one a rank-ordered list renders as simply the bottom row. An interval reported as zero with a stated denominator communicates it correctly, and the distinction matters: a product with a small measured share has a position to defend, whereas a product with none has not yet entered the consideration set at all.

The exact tie. In databases, Supabase and Neon recorded 217 and 216 mentions respectively — a difference of 1 mention across 500 answers, and an identical 10.3% share of model. One of them is printed above the other on any ranked list. That ordering is an artefact of tie-breaking, not a measurement, and it is exactly the kind of distinction a bare rank order manufactures.

4.4 Citation extraction is not yet reliable

An earlier draft of this paper reported two findings derived from citation rates: a weak positive correlation between share of model and citation rate ($r = +0.32$ across 90 products), and a wide categorical spread in citation rate (0.0% to 35.8%). While auditing the measurement pipeline we found that the extraction layer producing those rates does not work uniformly across engines. Both findings are withdrawn. What follows is what we found, reported as a result about the instrument.

Engine	Distinct cited domains per category	Across all 9 categories
Perplexity	5 (at the list cap of 5)	45
Google Gemini	1–2	13
ChatGPT (OpenAI)	5 (at the list cap of 5)	45
Claude (Anthropic)	5 (at the list cap of 5)	45
Grok (xAI)	0	0

Table 2. Distinct cited domains recovered per engine. The source list is capped at 5 entries, so 5 indicates saturation of the cap rather than a true count; values below it are real shortfalls.

Three failures are visible. First, **Grok (xAI) returned zero distinct cited domains in 9 of 9 categories** — not few, none. Second, **Google Gemini returned 1 to 2 distinct domains per category, 13 in total**, against a cap of 5 that Perplexity, ChatGPT (OpenAI) and Claude (Anthropic) each reached in every category. Third, of 34,801 extracted citations, **13,667 (39%) resolved to a platform redirect wrapper** rather than to a publisher domain.

Any per-product citation rate is therefore computed over five engines of which two contribute close to nothing, and over a pool in which roughly two of every five entries is not a source. The reported correlation is correct arithmetic on systematically understated inputs, and the categorical spread may reflect which engines happened to yield extractable citations in which categories rather than any property of the categories themselves. Neither can be interpreted, and we do not interpret them.

One observation from the same data is worth recording precisely because it is now ambiguous. 6 products hold a share of model of 10% or more while showing a citation rate of exactly 0.0% — including Prometheus at 11.3% and Jenkins at 16.6%. Under a working extractor this would be a substantive finding about engines naming products without sourcing them. Under this one it is indistinguishable from extraction failure. We report it as unresolved, and it is a question the corrected instrument is designed to answer.

We report this rather than quietly dropping the affected sections for two reasons. The defect is visible in the released data, so a reader would find it. And the same failure mode plausibly affects other published work: extraction quality is not reported anywhere we have looked, and cannot be inferred from a leaderboard.

5. Discussion

5.1 A rank without an interval overstates what was measured

If rank one and rank two are not separable in 6 of 9 categories, then for those categories the ordered list asserts something the data does not support. This is not a presentational nicety. A vendor told they are second may fund a programme to become first when no measurable gap exists, and may later attribute a change of position to that programme when the position was never stable.

The remedy is cheap. Publish the interval next to the estimate, state the number of runs the estimate rests on, and mark blocks of products that are not separable. None of this requires new measurement; it requires reporting what the measurement already contains.

5.2 Naming is measurable today; citing is not

Section 4.4 supports a claim we did not set out to make and now regard as the more consequential half of this paper. Whether an engine *names* a product is directly observable in the answer text, and with enough repetition it can be estimated with a defensible interval. Whether an engine *cites* a product's domain depends on an extraction step whose reliability varies by engine, is invisible to the reader, and in our measurement failed outright for one engine in every category.

Citation-share metrics are nonetheless published across this sector without any accompanying statement of extraction quality. Our results do not show that those figures are wrong. They show that a reader has no way to tell, and that at least one implementation — ours, which we can inspect — would have produced confidently stated figures that were substantially incomplete. Any index reporting citation share should report per-engine extraction coverage alongside it.

5.3 What we do not claim

These are 5 engines, 9 categories and one measurement window, and engine behaviour changes continuously. We make no claim that these rankings hold at any later date, that they generalise beyond B2B software, or that share of model predicts commercial outcomes. The claim is narrower and, we think, more durable: **whatever is measured should be reported with its uncertainty, and a measurement layer should be validated before results are drawn through it.**

6. Limitations

Extraction reliability. The citation-extraction layer failed for one engine in all 9 categories and under-recovered for a second, and 39% of extracted citations were redirect wrappers. Section 4.4 is a finding about this defect, not a result obtained despite it. Share-of-model results derive from brand naming and are unaffected.

Interval estimator. All intervals are Wilson score intervals. Share of model is a ratio over a pooled mention denominator whose observations are clustered within answers and within prompts, so the independence assumption is violated and the reported share intervals are narrower than a cluster-corrected estimate would give. Section 3.4 sets out why this makes the central finding a lower bound. A clustered bootstrap is the correction and is planned for the next measurement cycle.

API access, not consumer interfaces. Engines were queried through their APIs. Consumer products may apply different retrieval, ranking or personalisation, so these results approximate but do not reproduce what an individual user sees.

A single window. All measurements were taken on 2026-07-01. Longitudinal stability is unaddressed here and is the subject of separate work.

Ten prompts per category. Prompts were pre-registered and are released in full, but ten prompts cannot span a category's full buyer intent. A different set would likely shift shares somewhat.

Overlap as the separability test. Interval overlap is conservative and understates the number of genuinely significant differences. A formal difference test would separate some pairs we report as indistinguishable.

Naming is not endorsement. A mention counts a product as named in a recommendation context. It does not measure sentiment, and a product named as a caution counts the same as one named as a first choice.

Commercial interest. The author sells services related to the subject of this paper. The pre-registration, the released raw data and the published null result are the mitigations offered; readers should weigh them against the interest.

7. Data availability

All 9 category datasets are released under CC BY 4.0 as CSV and JSON, with per-product mentions, share of model, both confidence bounds, appearance rate and citation rate; the pre-registered prompt set and brand universe with their hashes; per-engine breakdowns; and the citation blocks **exactly as captured, including the redirect wrappers**. The defect described in Section 4.4 is therefore independently checkable rather than merely disclosed, and the released files are the ones the analysis in this paper was run against. The full-text answer corpus is retained and available for verification on request; it is not released in bulk.

8. Declaration of interest

The author is the founder of Clear Cited, which sells measurement and advisory services relating to brand visibility in AI answer engines. This work was self-funded and received no external support. No brand named in this paper was a client at the time of measurement, and no brand paid for inclusion, placement or review. The conflict-of-interest policy that governs the underlying index is published with the data.

9. Version history

v1.0 (26 July 2026) — first draft. Reported the share-of-model results together with two citation-level findings and a headline count of 34,801 citations.

v1.1 (27 July 2026) — current. Citation-level findings withdrawn following discovery of the extraction defect; the defect is reported as a result in Section 4.4; the citation count is removed from the abstract; Section 5.2 added; Limitations extended with extraction reliability and the interval estimator. Section 3.4 corrected: all intervals are Wilson, not a Wilson/bootstrap mix as v1.0 stated, and the consequence for the central finding is now stated. Share-of-model results are unchanged.

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